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Paleomagnetic and $^{40}Ar/^{39}Ar$ Studies of Late Kibaran Intrusives in Burundi, East Africa: Implications for Late Proterozoic Supercontinents[1]

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ABSTRACT

The Kibaran tectonothermal event in East Africa (1400–1200 Ma) peaked at 1300 Ma and was followed by the emplacement of a linear belt of mafic/ultramafic and felsic plutons between 1275–1220 Ma. A series of the mafic/ultramafic plutons in Burundi from the Congo craton carry a stable paleomagnetic remanence with a mean direction of $D = 108°$, $I = +16°$ ($N = 10$ sites, $k = 37$, $\alpha 95 = 8°$) with a corresponding paleomagnetic pole at 17° S, 113° E. Biotite samples from the Mukanda-Buhoro massif yield well-defined $^{40}Ar/^{39}Ar$ plateau ages averaging 1223 ± 4 Ma and slightly younger integrated ages of 1211 ± 2 Ma. The $^{40}Ar/^{39}Ar$ ages, combined with previously published U-Pb and Rb-Sr ages, suggest that the magnetic directions were acquired during cooling of the massif between 1260–1210 Ma (1236 ± 24 Ma). One pluton at Nyabikere yields an anomalous direction of $D = 46°$, $I = 29°$ and a corresponding virtual geomagnetic pole at 43° N, 137° E. Hornblendes and biotites from Nyabikere yield younger $^{40}Ar/^{39}Ar$ ages of c. 950 Ma. The magnetization and K-Ar ages in the Nyabikere rocks were reset during a c. 950 Ma thermal event known elsewhere in East Africa. Paleomagnetic poles for the interval 1200–1250 Ma from the Kalahari craton, Congo craton, Laurentia, Baltica, and Australia suggest that the Neoproterozoic supercontinent of Rodinia was not yet fully formed at 1200 Ma. Alternative, previously published supercontinental configurations are not supported by these new paleomagnetic data. Assuming the validity of Rodinia for Late Proterozoic times, this implies that our paleomagnetic results provide evidence for mid-Proterozoic plate movements leading to Grenville-aged collisions (c. 1100–1000 Ma) and the assembly of Rodinia.

Introduction

Recent proposals for Proterozoic supercontinents advocate the juxtaposition of the East Antarctic craton and eastern Australia (all directions refer to present-day coordinates) against the western margin of Laurentia (Moores 1991; Dalziel 1991). This so-called SWEAT hypothesis was subsequently modified to include Baltica, Siberia, India, Madagascar, the Amazonian and Rio Plata cratons of South America, and the Congo, Kalahari, and West African cratons of Africa, to form a supercontinent called Rodinia (figure 1*a*,*b*; Hoffman 1991; Dalziel 1992). The configurations of Hoffman (1991) and Dalziel (1992) are based on geologic similarities along 1300–1000 Ma orogenic belts. These Grenvillian-Kibaran orogenic belts are thought to represent the suture zones between cratons during the assembly of Rodinia. Other configurations for Proterozoic supercontinents have been proposed, most notably by Piper (1987; figure 1*c*). According to Piper (1987), a global compilation of paleomagnetic poles conform to a common apparent polar wander path when rotated to his configuration. Both Dalziel (1992) and Piper (1987) have quantified the rotations required to reconstruct their supercontinental configurations; therefore, their reconstructions can be tested paleomagnetically if suitably well-dated poles are available from different continents.

The 1400–1200 Ma Kibaran tectonothermal event in East Africa peaked at about 1300 Ma and is represented by the N-NE trending Kibaran Belt (Tack et al. 1994; Klerkx et al. 1987). This belt can be traced from the Zaire-Angola-Zambia border in the southwest to Uganda in the northeast (figure

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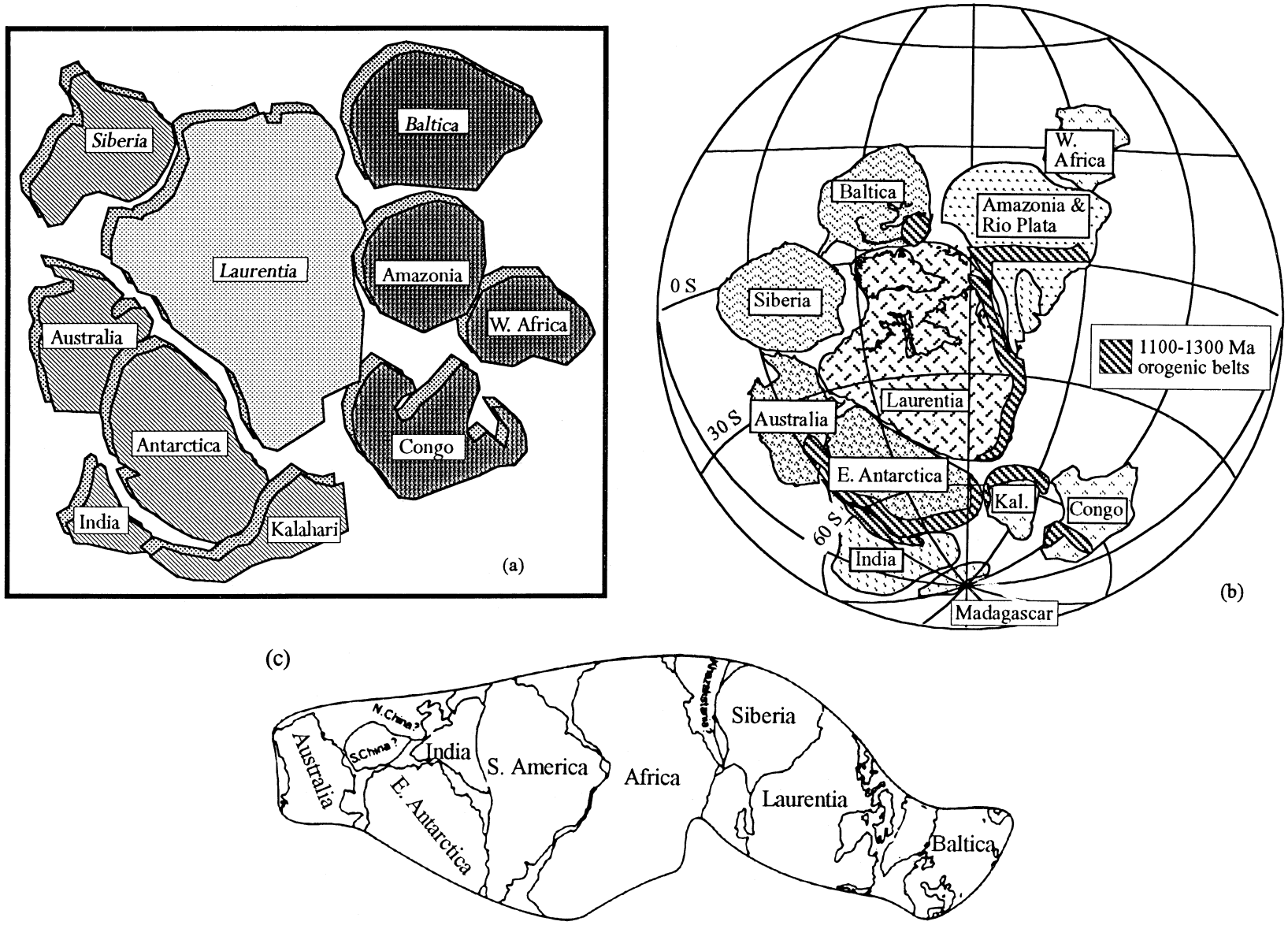


Figure 1. (*a*) The late Proterozoic supercontinent according to Hoffman (1991). (*b*) The Late Proterozoic supercontinent according to Dalziel (1992). The Grenvillian-Kibaran-Irumide age belts are shown for comparison. (*c*) The late Proterozoic supercontinent according to Piper (1987). The configuration in (*a*) is a qualitative reconstruction based on geologic similarities along contemporaneous orogenic belts of Grenvillian-Kibaran-Irumide ages. The configuration in (*b*) is a quantitative reconstruction fitting the orogenic belts and cratons on a constant radius earth. The reconstruction in (*c*) is based on an interpretation of the available paleomagnetic data.

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2). The rocks along this trend have received a variety of names, including Burundian, Karagwe-Ankolean, and Kibaran, but all were attributed to the same orogenic cycle. A series of late-Kibaran plutonic events are represented by bi-modal magmatic episodes in Burundi (Tack et al. 1990; Klerkx et al. 1987). These two episodes, one mafic-ultramafic and the other granitoid, are thought to coincide with the culmination of the Kibaran orogeny (Tack et al. 1994; Klerkx et al. 1987). Ages from these plutons range from 1275 to 1220 Ma (Tack et al. 1994; Liégois pers. comm. 1993; Tack et al. 1990; Klerkx et al. 1987). These intrusive events are broadly coeval with well-dated magmatic episodes in the 1250 Ma Gardar Province of Greenland (Paslick et al. 1993), the c. 1250 Ma Late Jotnian episode of Fennoscandia (Pesonen et al. 1989) and the c. 1200 Ma Premier Kimberlite intrusions of the Kalahari craton (Dopplehammer and Hargraves 1994).

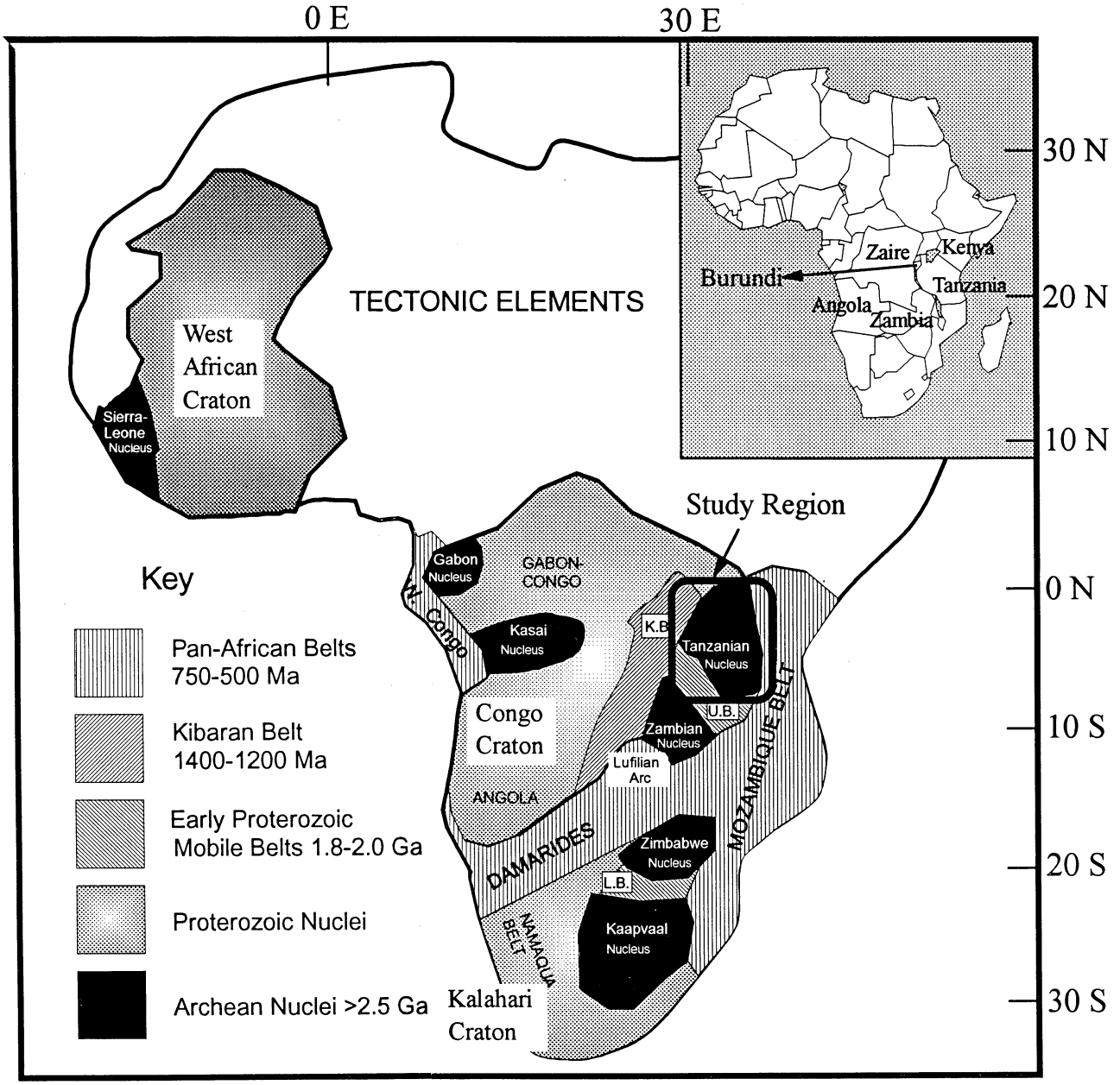


Figure 2. Tectonic elements of the African continent and their estimated age ranges. (K.B. = Kibaran Belt; U.B. = Ubendian Belt; L.B. = Limpopo Belt).

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The exact positioning of the Congo craton within Rodinia appears uncertain. Hoffman (1991; figure 1*a*) places the Congo craton adjacent to southern Laurentia maintaining a "loose" fit of the 1100–1300 Ma Grenville-Kibaran-Irumide orogenic belts. Dalziel (1992; figure 1*b*) places the Congo craton adjacent to the Kalahari craton in their present-day relative positions. Because the 1200–1250 Ma paleopositions of Laurentia, Baltica, and the Kalahari craton are well-constrained by paleomagnetic poles from the above-mentioned regions, a well-dated pole from the Kibaran plutons of the Congo craton can provide an excellent test of the continental configurations at c. 1250 Ma. A paleomagnetic pole of this age might also shed light on the ensialic/ensimatic nature of the younger orogenic belts (figure 2) separating the Congo and Kalahari cratons.

Geologic History

The Kibaran tectonothermal event has been interpreted by Klerkx et al. (1987) as a polyphase orogenic cycle of intracratonic (ensialic) deformation. The exact nature of the Kibaran orogenic event is still debated (Klerkx et al. 1993; Rumvegeri 1991; Kröner 1977). Rumvegeri (1991) suggests that the succession of tectonic, metamorphic, and magmatic features are indicative of subduction/collision between the Gabon-Congo and Tanzanian cratons, but both Kröner (1977) and Klerkx et al. (1987, 1993) interpret the tectonometamorphic event as entirely intracratonic. There is, however, general agreement on the timing of tectonic and metamorphic events during Kibaran orogenesis. The first phase of deformation began at c. 1330 Ma and is characterized by amphibolite facies metamorphism and emplacement of pre- and syntectonic granites and granodiorites (Rumvegeri 1991; Klerkx et al. 1987). This was followed by the main Kibaran deformation at 1300 Ma accompanied by granulite facies metamorphism and subsequent 1220–1280 Ma bimodal plutonism (figure 3; Tack et al. 1994). The final Kibaran involved late shearing during local greenschist metamorphism at ≈1150 Ma (Tack et al. 1994). There are two other magmatic associations in the Kibaran Belt that followed the main orogenic episodes. These are emplacement of tin-bearing granites in NE Burundi, NW Tanzania, and SW Zaire between 900–1000 Ma (Cahen et al. 1984; Ikungura et al. 1992) and minor Pan-African intrusions at 750 Ma (Tack et al. 1984).

Paleomagnetic Studies

Sampling and Results. Figure 3 shows the locations of the 11 paleomagnetic sampling sites in Burundi. Six to seven 25 mm diameter cores were drilled at each site and oriented with sun and magnetic compass. These cores were cut to specimens ≈22 mm in length for measurement in the paleomagnetic laboratory at the University of Michigan.

The intensities of the natural remanent magnetization (NRM) range between 10^{-4} and 1 A/m as measured on either an ScT cryogenic magnetometer or a Schonstedt spinner magnetometer. Thermal and alternating field (AF) demagnetization carried out on a selection of pilot samples indicated that AF demagnetization would result in a clearer definition of the magnetic components (figure 4). Demagnetization results were plotted in orthogonal vector diagrams (Zijderveld 1967), and linear trajectories were used to determine directions of magnetic components through the use of principal component analysis (Kirschvink 1980). Most samples showed a distribution of points that follow great-circle trajectories (figures 5–7). For these samples, great-circle analysis was combined with stable endpoint data to determine site mean directions following the methods of Halls (1978) and Bailey and Halls (1984). Samples often showed great-circle trajectories reversing from one polarity to the opposite polarity (e.g., sample b91-202, figure 5).

A total of 55 samples (out of 72) gave stable directions or easily interpretable great-circle trajectories (table 1). The overall mean direction for the Burundi plutons was determined by combining the means of stable endpoint directions and great-circle intersections (figure 8). The mean direction calculated using only great-circle intersections is $D = 106°$, $I = +15°$ ($N = 9$ sites; $k = 30$, $\alpha 95 = 9°$). The mean direction calculated using only stable endpoint directions is $D = 113°$, $I = +13°$ ($N = 7$ sites; $k = 139$, $\alpha 95 = 5°$). The combined mean direction using both great-circles and stable endpoints is $D = 108°$, $I = +16°$ ($N = 10$ sites; $k = 37$, $\alpha 95 = 8°$). The resultant paleomagnetic pole is 17° S, 113° E ($K = 51$, A95 = 7°). Because of the consistency in paleomagnetic directions from site-to-site and across shear zones, we argue that the post-metamorphic plutons have not been subjected to any significant tilting or rotation since cooling through their magnetic blocking temperatures. Site 11 at Nyabikere showed a different direction from the other sites and was not included when calculating our mean direction for the late Kibaran intrusives. These four samples yield a mean of $D =$

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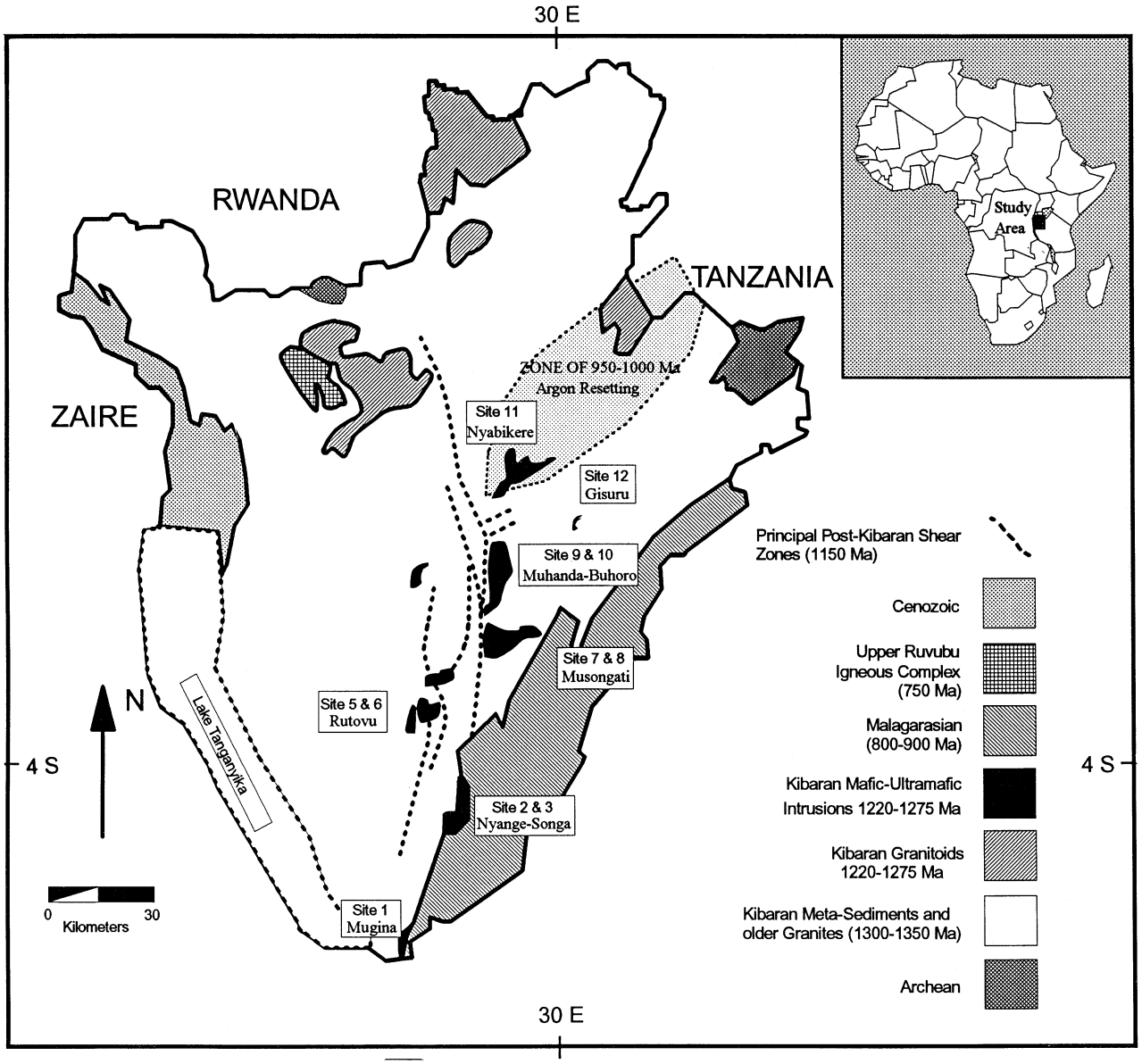


Figure 3. Generalized geologic map of Burundi showing the locations of the mafic/ultramafic plutons where paleomagnetic samples were obtained. An outline of the region in Burundi and Tanzania where resetting of argon ages and paleomagnetic directions occurred during the intrusion of tin-bearing granites between 950–1000 Ma is shown in light gray shading.

46°, $I = -29°$ ($k = 25$, $\alpha 95 = 14°$). The calculated virtual geomagnetic pole for site 11 is 43°N, 137° E. This deviating result will be discussed in more detail below.

Rock Magnetic Experiments. Samples were subjected to a number of rock magnetic tests to determine the carrier of their magnetic remanence. Isothermal remanence acquisition (IRM) studies show a range of behavior with two of the samples reaching saturation by 0.2 Tesla (e.g., samples b91-1002 and b91-703 in figure 9*a*), suggesting that magnetite is the primary carrier of the magnetic remanence in those samples. This observation agrees with the coercivity spectra during AF demagnetization and the unblocking temperature spectra. However, because other IRM plots show higher inductions to reach saturation, which may indicate hematite or pyrrhotite, other rock magnetic experiments were conducted to verify the carrier of magnetic remanence. Johnson et al. (1975) proposed a modification of the Lowrie-Fuller (1971) test whereby the AF stability of IRM remanence is compared to ARM remanence. These tests show that the AF coercivity of the 0.1 T ARM is stronger

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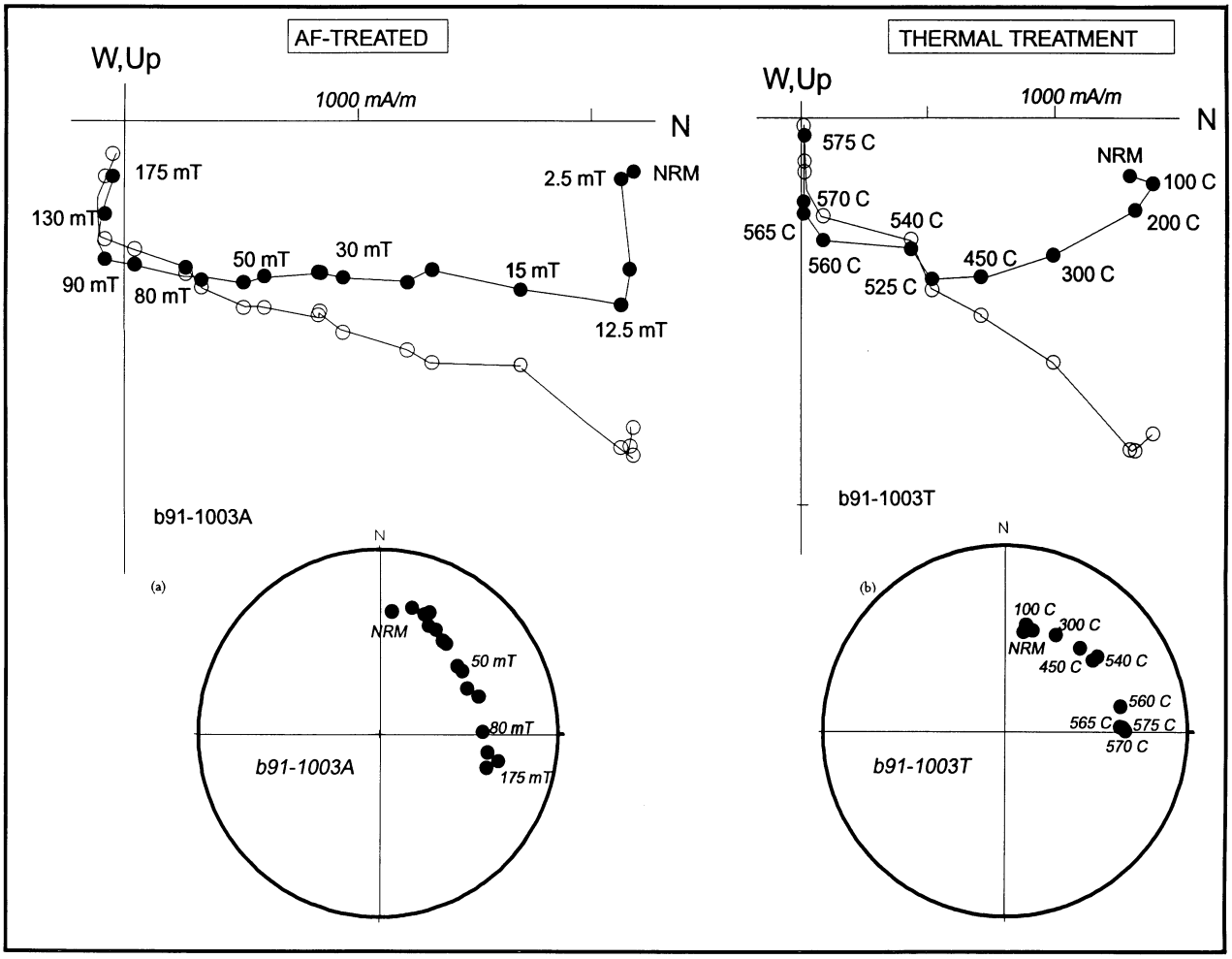


Figure 4. Results of alternating field (AF) and thermal treatment on sample b91-1003 (Mukanda-Buhoro). Open circles represent projections on the vertical plane and closed circles represent projections on the horizontal plane. Although both treatments yield similar results, alternating field produced a clearer demagnetization trajectory in most samples.

than the AF coercivity of a 0.1 T IRM, indicative of single-domain or pseudo single domain grains (figure 9*a,b*). Magnetite is also indicated by Curie temperature experiments (figure 10*a*) and thermal demagnetization of a 3-orthogonal axis IRM (figure 10*c,d;* Lowrie 1990). Site 7 (figure 10*d*) also shows a significant contribution from pyrrhotite that is demagnetized at 280° C. Site 12 shows a contribution from hematite or pyrrhotite (figure 10*b*) that is also partially reflected in the IRM curves (figure 9*a*) We conclude that the likely carriers of remanence in these rocks are magnetite with some contribution of hematite and pyrrhotite.

Magnetic Petrography. The samples reported on here range from gabbro (Sites 1, 6, and 12), leucogabbro (Sites 2, 3, and 5), to norites or leuconorite (Sites 7, 8, 9, and 10). Site 11 is in an ultramafic pyroxenite. Except for the last (Site 11, the only sites that yielded anomalous directions) the rocks are very fresh, with only minor alteration of mafic minerals or feldspar. The leuconorites from Mukanda-Buhoro (Sites 9 and 10) contain fresh labrodorite and bytwonite plagioclase feldspars, large oikocrysts of orthopyroxene with conspicuous clinopyroxene exsolution lamellae and overgrowths. With respect to opaque phases, the gabbros appear to have laths of ilmenite as the only primary accessory. The leucogabbros also contain primary exsolved titaniferous magnetite intergrowths, in which the coarse ilmenite lamellae are completely fresh, but the magnetite host is extensively altered or oxidized. In the norites and leuconorites, pyrite is the predominant accessory with relatively minor ilmenite. A drill core, supplied to us by Luc Tack of the Musee Royal de L'Afrique Centrale, from late pegmatitic veins in the Mukanda-Buhoro mas-

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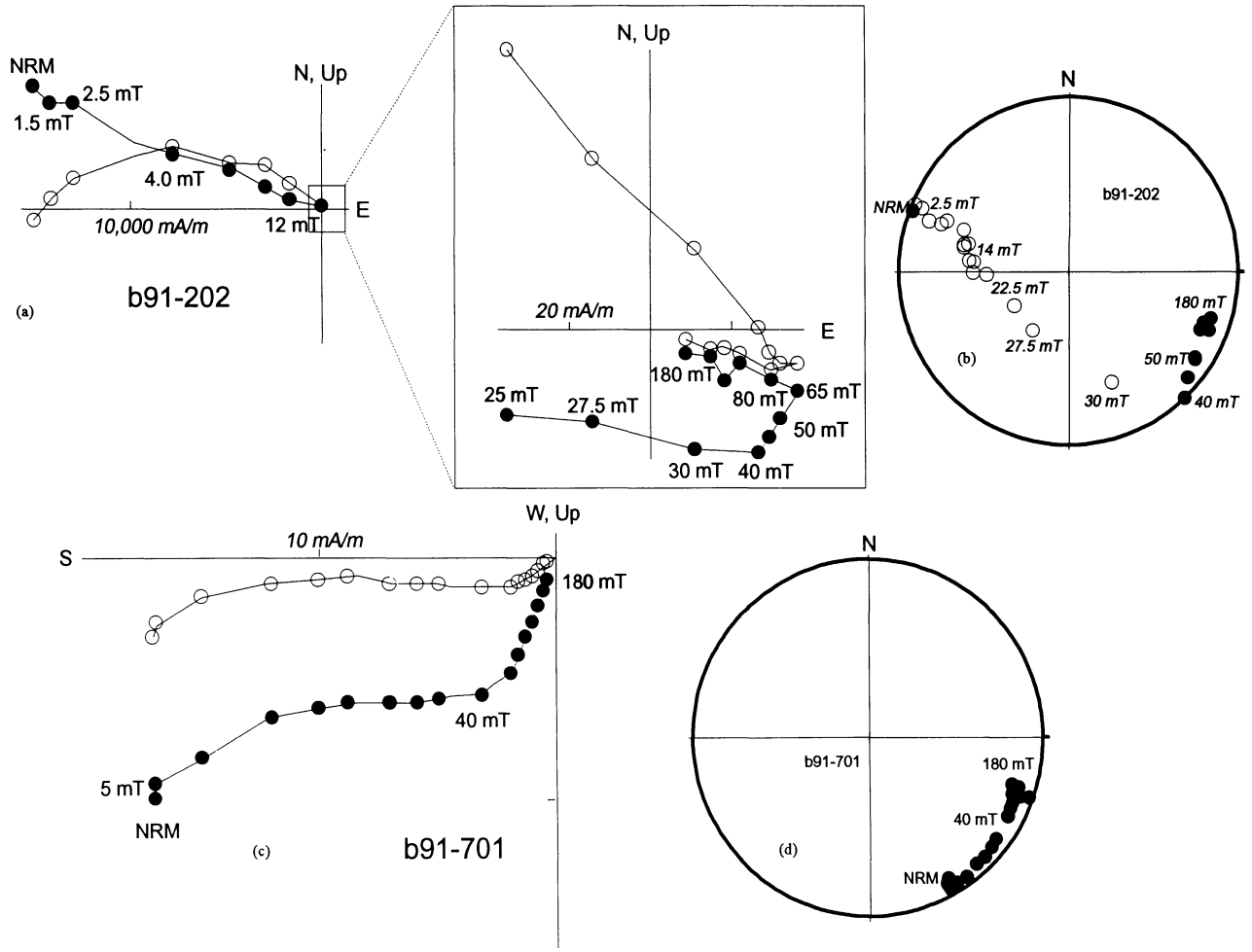


Figure 5. Typical AF demagnetization behavior for samples from the late Kibaran intrusives (orthogonal vector plot symbols as in figure 4). (*a*) sample b91-202 (Nyange-Songa) shows great-circle trajectory from one polarity to the other (*b*) equal-area stereographic projection for sample b91-202 (open symbols on stereoplots are upper hemisphere projections and closed symbols are lower hemisphere projections). (*c*) Orthogonal vector diagram for sample b91-701 (Musongati) and (*d*) equal-area stereographic projection of sample b91-701 (symbols as in 5*b*).

sif yielded 0.1 to 0.2 centimeter-sized mica grains used for $^{40}Ar/^{39}Ar$ dating.

At Nyabikere, the rocks are extremely altered. The original composition resembles that of a feldspathic pyroxenite, but the pyroxene is now nearly completely replaced by needles of chlorite. Biotite appears to be partly replaced by chlorite and occurs around grains of ilmenite. Minor amounts of hornblende appear to replace pyroxene in some of the cores examined. Both biotite and hornblende were separated from the Nyabikere rocks for $^{40}Ar/^{39}Ar$ dating.

The most conspicuous feature of these rocks is the scarcity of primary magnetite. Ilmenite and/or pyrite ± pyrrhotite are the predominant opaque phases in most samples. In those which originally contained titaniferous magnetite, the magnetite host to the intergrowth is pervasively oxidized or altered. By default, the only magnetite present must occur as very fine-grained particles either exsolved from the silicates or produced by deuteric alteration. This exclusively fine-grained size of the magnetite and/or hematite must account for the stability of the remanent magnetization in these rocks. These petrographic observations are consistent with the results of the rock magnetic experiments that indicate the remanence is carried by single-domain to pseudo-single domain magnetite with some contribution from hematite and pyrrhotite.

Age of the Paleomagnetic Pole

The mafic/ultramafic intrusions in Burundi are interpreted to be broadly cogenetic with the granitoid intrusions on the basis of their structural asso-

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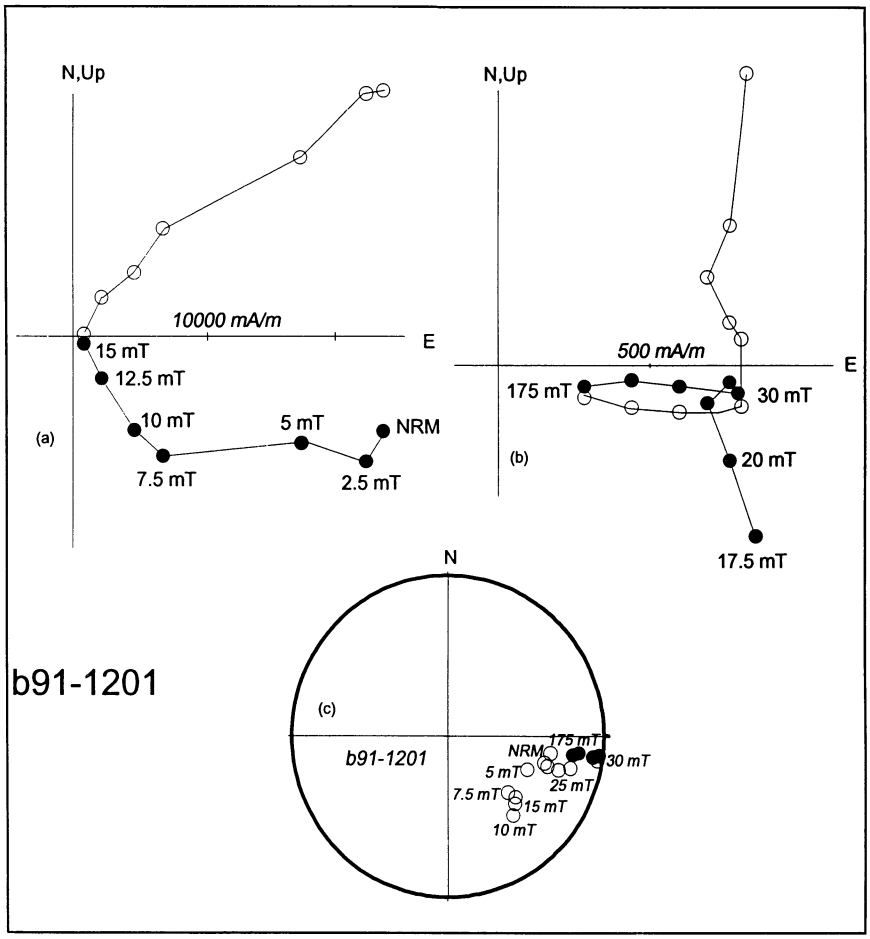


Figure 6. (*a*) Orthogonal vector diagram with blowup (*b*) and equal-area stereographic projection (*c*) for sample b91-1201 (Gisuru; symbols as in figures 4 and 5) showing the characteristic direction isolated between 30-175 mT.

ciations with the late Kibaran shear zones, similar age determinations, and the observation that both rock types appear intimately intermixed within a single massif. The granitoids are thereby interpreted as anatectites or differentiation products of the mafic/ultramafic rocks (Tack et al. 1990, 1994). The granitoid differentiate of the Musongati massif yielded a U-Pb zircon age of 1275 +11/−10 Ma (Tack et al. 1994). A U-Pb zircon age of 1249 +8/−7 Ma was obtained from the Bukirasazi granite and a number of $^{207}Pb/^{206}Pb$ ages in the range of 1220–1275 Ma were derived from the Musongati massif (Tack et al. 1994; Liégois et al. pers. comm. 1993). A less well defined Rb-Sr 4 point isochron on calcite-mica pairs from samples of the Mukanda-Buhoro drillcore yielded an age of 1236 ± 70 Ma (Tack et al. 1990). Additional Rb-Sr ages from these intrusions were between 1090 and 1137 Ma (Klerkx et al. 1987). The younger ages are believed to reflect Rb-Sr rehomogenization during the late Kibaran shearing event (Tack et al. 1994). Because of the uncertainty regarding the exact age of the magnetization, rock samples from the Mukanda-Buhoro massif (paleomagnetic sites 9 and 10), and Nyabikere (paleomagnetic site 11) massif were analyzed using the $^{40}Ar/^{39}Ar$ dating method.

Methods. Rock samples were crushed, sieved, and the individual grains of biotite and hornblende were separated, purified, and hand-picked for analysis. After separation, the samples were washed in water and acetone, then rinsed numerous times in de-ionized water. The samples were packaged in pure aluminum foil containers, sealed in evacuated fused silica vials and irradiated at the University of Michigan's Ford Phoenix Reactor at location L67 for 90 MWh. Following irradiation, the samples were stepwise treated using an argon-ion laser system. The argon isotopic ratios of masses 40–36 were measured on a Mass Analyzer Products 215 mass spectrometer operating in static mode. The ion source was operated at 100 μA trap current with an approximate source efficiency of 3×10^{-4} A/torr. This mode of operation optimizes signal/background ratios of mass 36 using the electron multiplier. The mass spectrometer is equipped with two side-by-side detectors, a Faraday cup and a 17-stage Balzers electron multiplier selectable under computer control. Most of the analyses were

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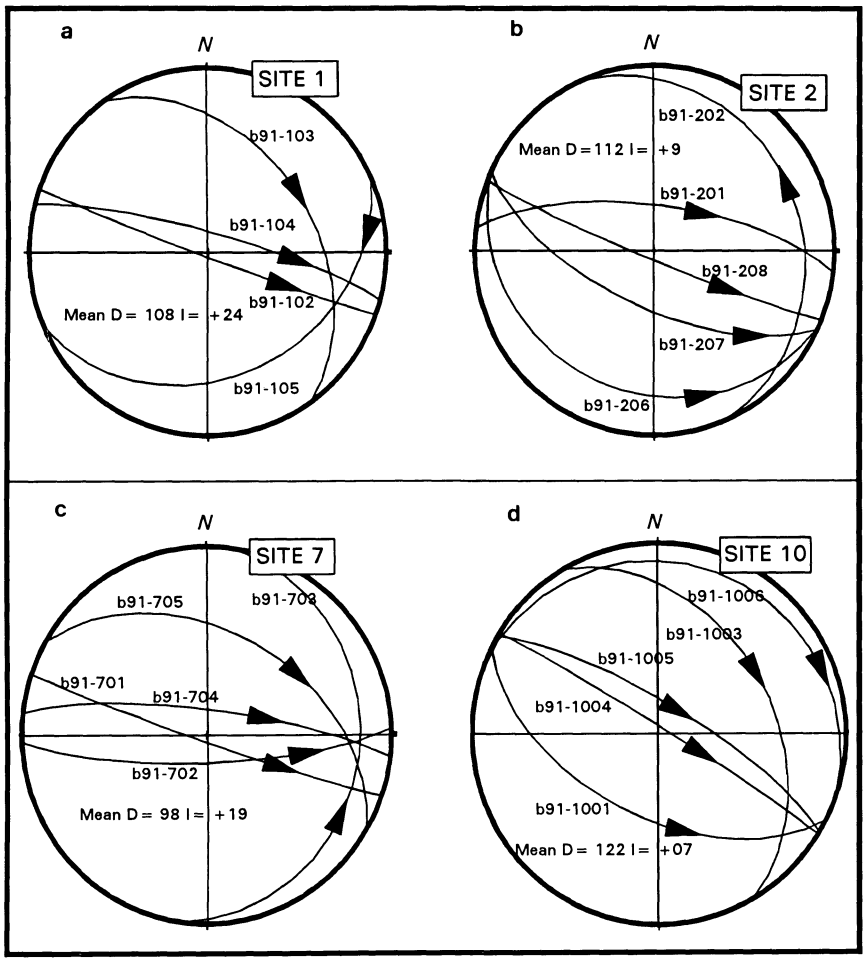


Figure 7. Lower hemisphere equal-area stereographic projection for several sites showing sample great-circle trajectories toward the characteristic direction. (*a*) Site 1 = Mugina; (*b*) Site 2 = Nyange-Songa; (*c*) Site 7 = Musongati and (*d*) Site 10 = Mukanda-Buhoro.

performed entirely using the electron multiplier; however if a signal was off-scale for the electron multiplier, it was measured with the Faraday detector. In this latter case the multiplier gain is monitored either during the run using a suitable isotope measured by both detectors, or at the end of the analysis by partially pumping out the gas to bring the large peak back on scale. Mass discrimination was monitored daily. To measure the discrimination in the source with the Faraday collector, approximately 2.6×10^{-7} cc STP of Ar was measured. The $^{38}Ar/^{36}A$ ratios would then be used to estimate source plus multiplier mass discrimination. More typically, a small percentage of an air pipette's volume of $\approx 3 \times 10^{-9}$ cc STP would be leaked into the mass spectrometer, and the resulting fractionated $^{40}Ar/^{36}Ar$ ratio was used to correct for source plus multiplier mass discrimination. Step-heating was accomplished by increasing the laser power with the laser beam defocused to cover the entire sample. Each step consisted of heating for 60 seconds followed by 2 minutes of gas purification using two 10 l/s SAES getters (ST101 alloy) and a liquid N_2 cold finger. Neutron flux gradients were monitored with the MMhb-1 standard (Alexander et al. 1978) using an age of 520.4 Ma. Samples consisted of 6–10 grains of hornblende or single grains of biotite. In general, two or three replicates, or splits, of each sample were run to determine any grain-to-grain variability. Extraction line blanks were run every five fractions and peaks are all blank corrected. Analytical data for the individual samples/splits are given in table 2.

Results. Figure 11 shows typical argon release spectra for single grains of biotite from the Mukanda-Buhoro massif. Total gas ages are similar to the calculated plateau ages for these samples. We define a plateau when a sample released >50% of the gas over three or more successive steps with overlapping ages at the 2σ level and a mean square weighted deviates (MSWD) cutoff value of ≤ 2.0. All errors reported are 1σ unless otherwise noted. Plateau ages range from 1226 ± 4 Ma (MSWD = 0.32) to 1220 ± 2 Ma (MSWD = 1.2). In cases where the age spectra did not yield a plateau be-

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Table 1. Paleomagnetic Results for the Late Kibaran Intrusive Rocks

Site #[a]	n/N[c]	Dse[d]	Ise[e]	Dcg[f]	Igc[g]	D[h]	I[i]	k[j]	α95[k]	Plat[l]	Plong[m]
1-Mugina	4/6	. . .	. . .	108°	+24 (4)	108°	+24°	123	8°	18° S	108° E
2-Nyange Songa	8/8	113°	+9 (3)	110°	+9 (5)	112°	+9	43	11°	22° S	117° E
3-Nyange Songa	3/7	. . .	. . .	103°	+15° (3)	103°	+15°	26	14°	13° S	113° E
5-Rutovu	5/6	120°	+16° (2)	110°	+12° (4)	120°	+15°	51	9°	30° S	114° E
6-Rutovu	4/6	. . .	. . .	83°	+37° (4)	83°	+37°	24	18°	5° N	99° E
7-Musongati	7/7	115°	+15° (4)	98°	+19° (5)	105°	+16°	40	14°	15° S	113° E
8-Musongati	2/7	112°	+18° (2)	. . .	. . .	112°	+18°	150	. . .	22° S	112° E
9-Mukanda Buhoro	6/6	105°	+19° (3)	111°	+10° (5)	105°	+16°	18	14°	15° S	113° E
10-Mukanda Buhoro	6/6	115°	+9° (2)	122°	+7° (5)	119°	+8°	33	13°	29° S	118° E
11-Nyabikere[b]	4/7	47°	−30° (2)	42°	−27° (3)	46°	−29°	25	14°	43° N	137° E
12-Gisuru	6/6	109°	+6° (2)	104°	+00° (4)	106°	+3°	10	21°	16° S	120° E
Means		113° k = 139	+13° α95 = 5°	106° k = 30	+15° α95 = 9°	108° k = 37	+16° α95 = 8°			17°S K = 51	112.7° E $A95$ = 7°

Notes. Means are calculated by averaging individual SITE results. In some cases a sample was used to calculate both stable endpoint and great circle means. However, the samples were given only unit weight to calculate the combined mean. The numbers in (parentheses) reflect the number of samples used to calculate the mean. K = Fisher precision parameter for mean of VGP's; A95 = Cone of 95% confidence for mean of VGP's.

[a] Site Number corresponding to figure 3.
[b] Not used to calculate overall mean statistics [Nyabikere massif].
[c] Number of samples used/number of samples per site.
[d] Mean declination from stable endpoints.
[e] Mean inclination from stable endpoints.
[f] Mean declination from great circles intersections.
[g] Mean inclination from great circle intersections.
[h] Mean declination from combined stable endpoints and great circle intersection.
[i] Mean inclination from combined stable endpoints and great circle intersections.
[j] Fisher's precision parameter.
[k] Cone of 95% confidence about the mean direction.

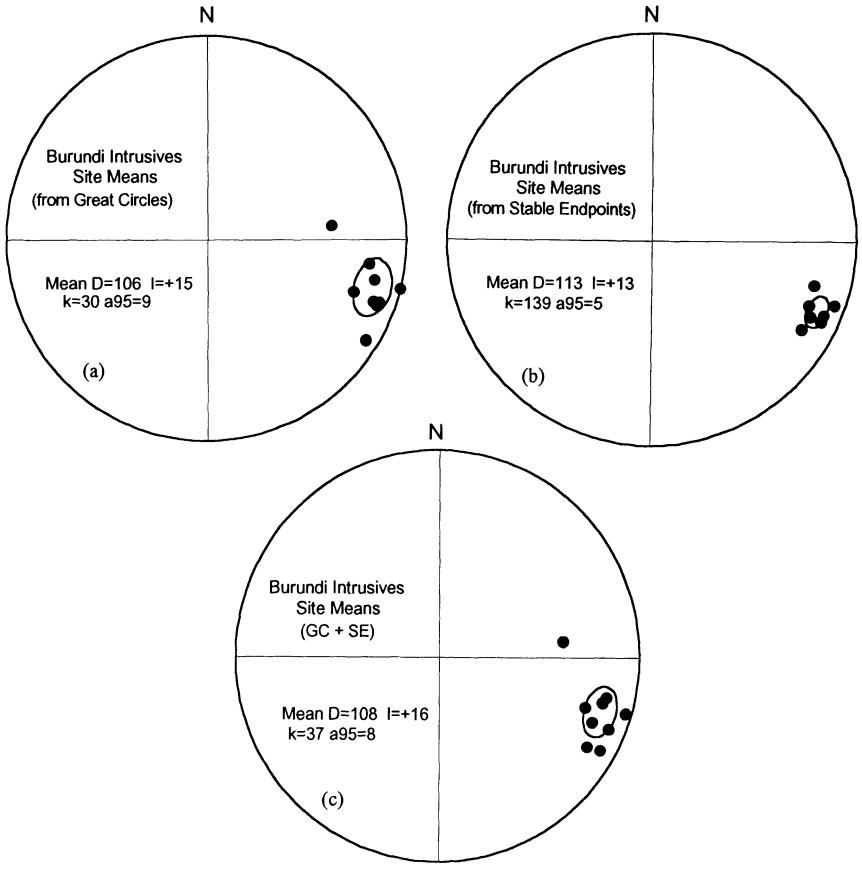


Figure 8. (*a*) Lower hemisphere equal-area projection of site means calculated using great-circle intersections; (*b*) lower hemisphere equal-area projection of site means calculated from stable endpoint directions; and (*c*) Site means calculated from combined great-circle and stable endpoint directions (note only nine sites visible because Sites 7 and 9 have identical means).

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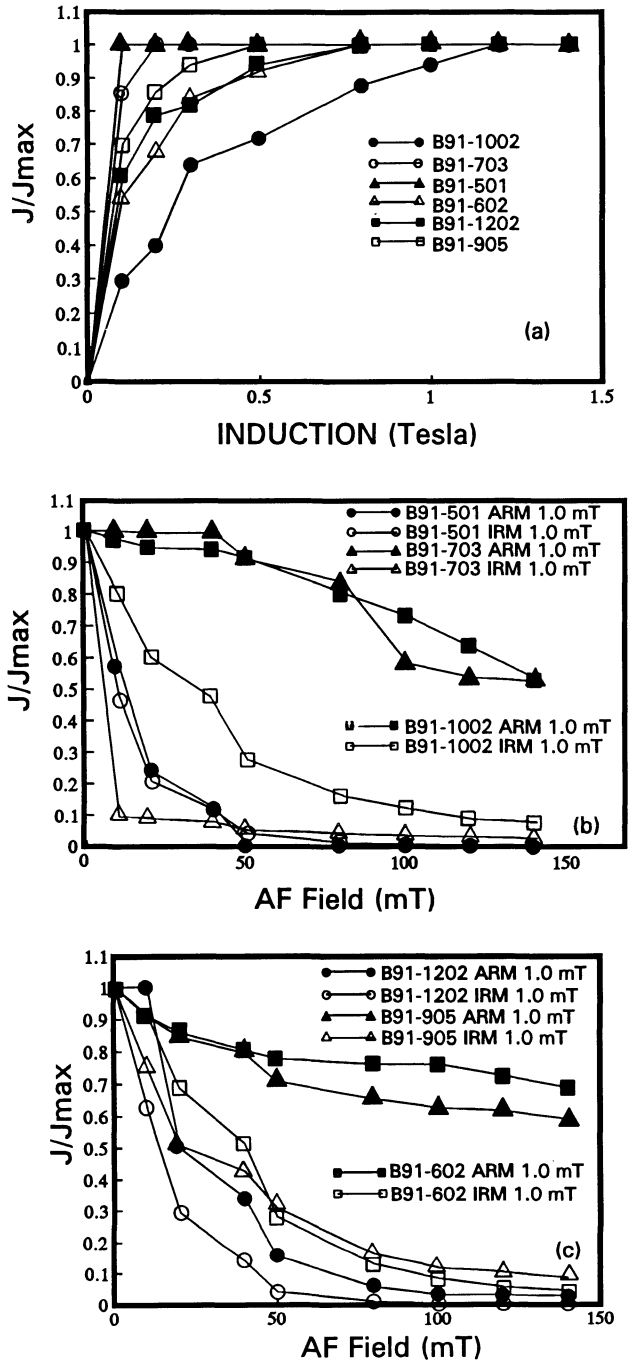


Figure 9. (*a*) Isothermal remanence acquisition (IRM) curves for samples from the late Kibaran intrusives; (*b,c*) response of paleomagnetic samples with IRM (0.1T) and anhysteretic remanent magnetization (ARM; 0.1T) to alternating field demagnetization indicative of single to pseudo-single domain grains (after Johnson et al. 1975).

cause the MSWD values were too high, we added all the gas fractions over the imperfect plateau region to arrive at an integrated age. Two splits of the mica yielded integrated ages over 80% of the gas of 1214 ± 3 (MSWD = 2.1) and 1212 ± 4 (MSWD = 3.7). Plateau ages can be interpreted in several different ways (McDougall and Harrison 1988). Minerals showing flat age spectra may reflect fast cooling of the mineral through the argon blocking temperature, secondary growth of the minerals due to a later thermal/chemical event, or that the diffusional characteristics of the mineral species are different in the laboratory than in nature.

Because the plateau ages recorded in the Mukanda-Buhoro rocks are: (1) similar to U-Pb crystallization ages found elsewhere in the related plutons; (2) correspond closely to Rb-Sr ages from the same rock samples; (3) are associated with a late hydrothermal event within the pluton; and (4) are older than the late Kibaran shear event, we conclude that the most reasonable interpretation of the age spectra is that the biotites cooled rapidly through their blocking temperature. The paleomagnetic samples exhibited a wide range of unblocking temperatures and coercivities. Several samples (e.g., Sample b91-202, figure 5) showed great-circle trajectories trending from one polarity to the opposite polarity and may have cooled through a field reversal. The magnetic carriers of the stable remanence observed in these rocks are believed to be single domain or pseudo-single domain magnetite grains with some contributions from hematite and pyrrhotite. The magnetite and hematite(?) grains were very likely produced at temperatures above their Curie points as fine grained exsolved particles in silicates or by high temperature deuteric alteration. It is thus likely that the magnetic remanence was acquired by these grains during the protracted cooling of the intrusions. The age of magnetization must therefore lie between 1275 Ma U-Pb age and the minimum 1211 Ma $^{40}Ar/^{39}Ar$ mineral age from Mukanda-Buhoro. We used a U-Pb zircon blocking temperature of ≈700°C (Harrison and McDougall 1980) and a $^{40}Ar/^{39}Ar$ biotite blocking temperature of ≈450°C (Giletti 1974) to estimate a nearly linear cooling rate for these massifs between 1275 and 1211 Ma. Because the magnetic blocking temperature ranged between 580° and 400°C, a reasonable estimate for the magnetization age is between 1211 and 1260 Ma (i.e., 1236 ± 24 Ma).

The intrusion at Nyabikere is different in terms of its paleomagnetic direction (table 2), petrogra-

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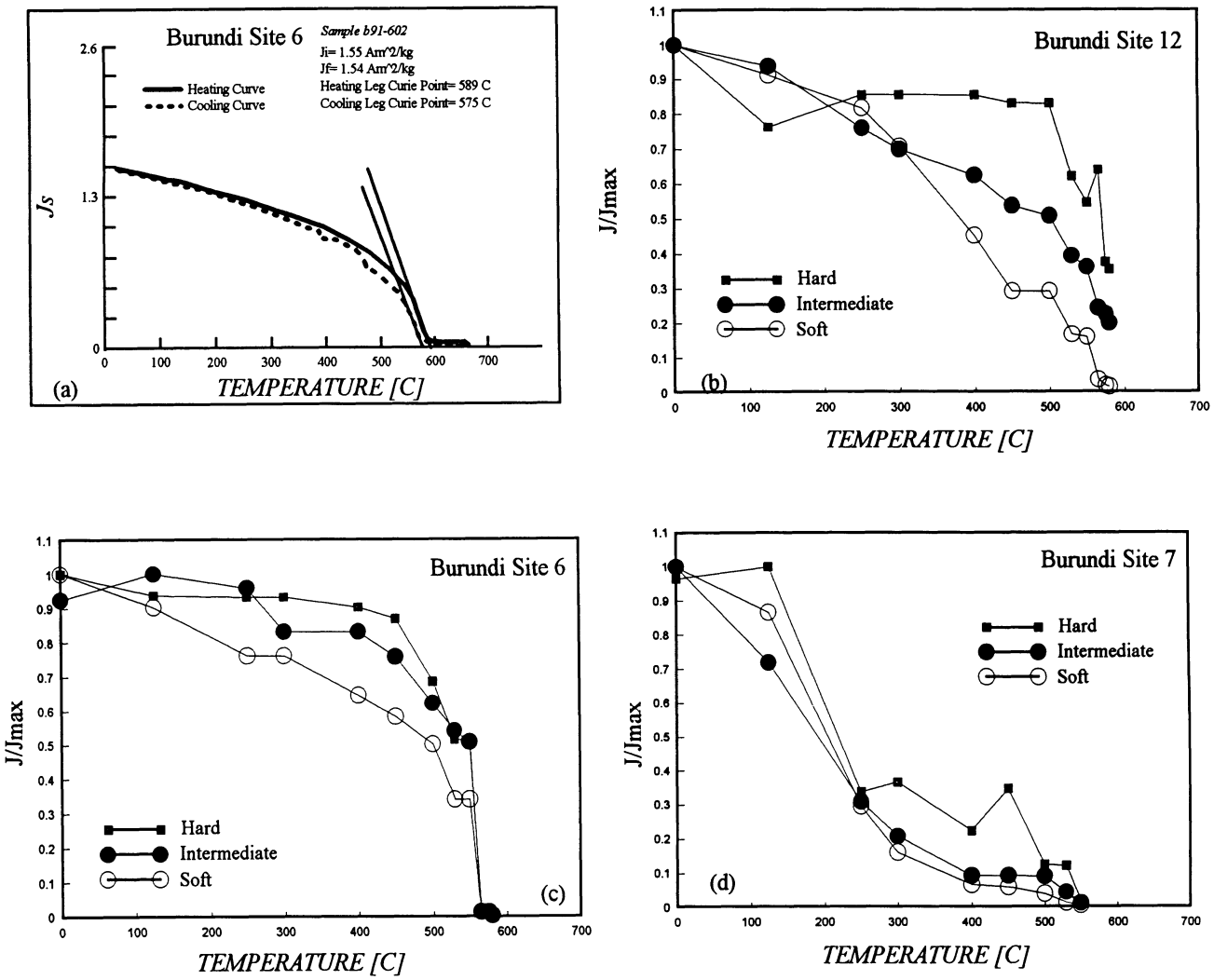


Figure 10. (*a*) Curie temperature run on sample b91-602 (Rutovu) showing Curie temperatures between 575–589°C indicating magnetite; (*b,c,d*) J/Jsat plots of thermal demagnetization of 3 orthogonal axis IRM after Lowrie (1990), indicating contributions from (*b*) magnetite and hematite; (c) magnetite; and (*d*) pyrrhotite and magnetite.

Table 2. $^{40}Ar/^{39}Ar$ Data Summary

Sample	J-Value ± Error	Total Gas Age ± 1σ (Ma)	Plateau (P) or Integrated (I) Age ± 1σ (Ma)	MSWD	Gas For (P) or (I) Age (%)
MUH-1 (Biotite)	.007476 ± .00014	1232 ± 8	1226 ± 4(P)	.32	73
MUH-2 (Biotite)	.007476 ± .00014	1212 ± 4	1214 ± 3(I)	2.10	99
MUH-3	.007348 ± .00013	1218 ± 2	1220 ± 3(P)	1.20	97
MUH-4 (Biotite)	.007348 ± .00013	1211 ± 2	1212 ± 4(I)	3.70	96
NYABIK hb1 (hornblende)	.007446 ± .00013	1042 ± 2	1004 ± 7(I) 920 ± 20[a]	30 1.3	66
NYABIK hb2 (hornblende)	.007282 ± .00014	1029 ± 3	1000 ± 3(P) 970 ± 8[a]	3.00 1.4	55
NYABIK bt1 (biotite)	.007203 ± .00011	891 ± 3	908 ± 2(P)	2.20	69
NYABIK bt2	.007203 ± .00011	931 ± 2	944 ± 3(P)	2.50	70

[a] Isochron age.

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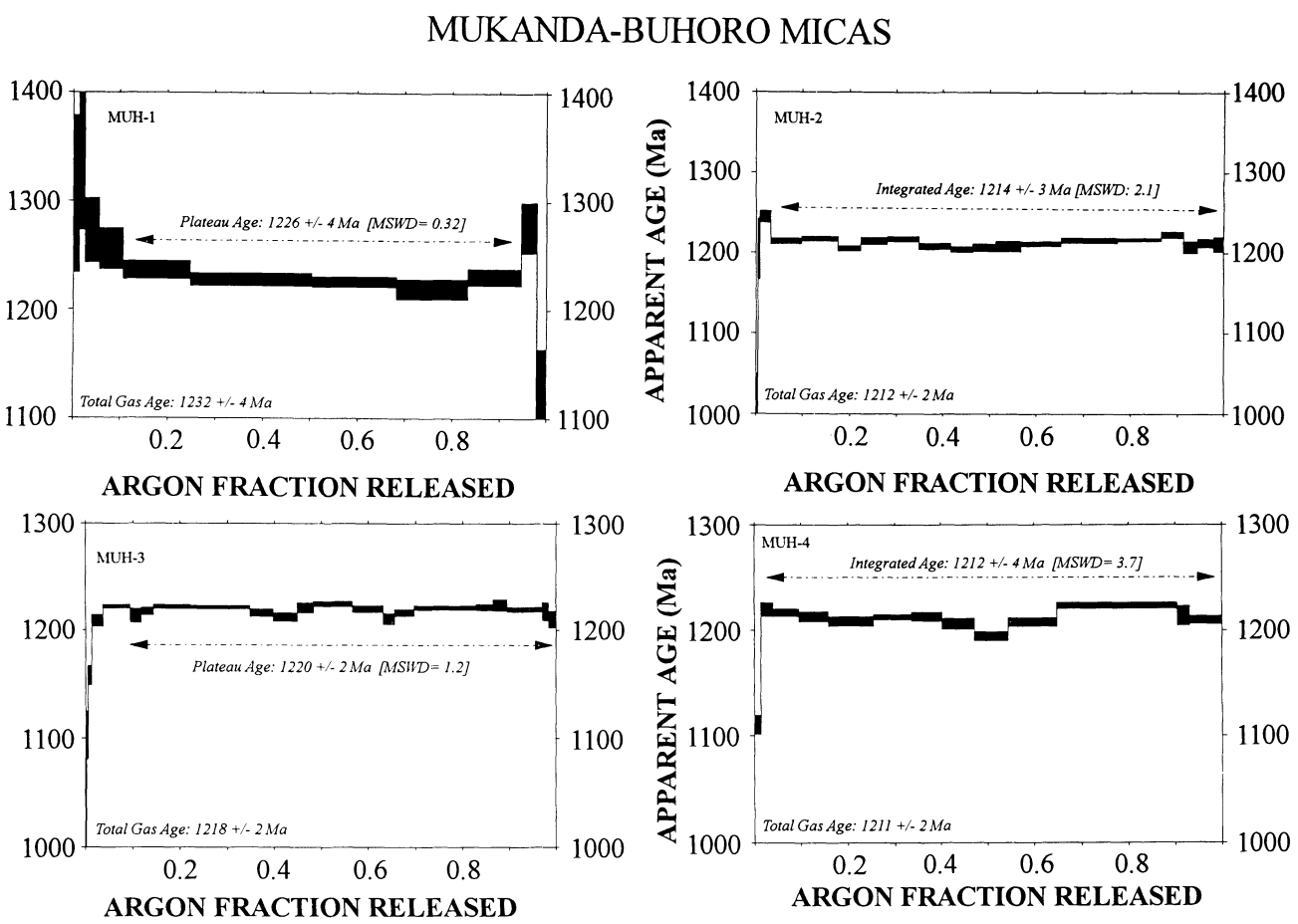


Figure 11. Typical argon release spectra yielding apparent ages for individual biotite grains from the Mukanda Buhoro massif.

phy, and age. It (figure 3) is close to several post-Kibaran tin-bearing granite intrusions described by Ikungura et al. (1992). This suite of post-Kibaran granites in NW Tanzania show disturbed $^{40}Ar/^{39}Ar$ spectra for biotites which yielded ages between 900 and 950 Ma (Ikungura et al. 1992). We analyzed hornblendes and biotites from a single sample of the Nyabikere massif. The hornblendes exhibit disturbed age spectra yielding ages of about 1000 Ma (figure 12*a*,*b*). Isochron ages for these same samples yield ages of 920 ± 40 Ma (MSWD = 1.3; sample in figure 12*a*) and 970 ± 16 Ma (MSWD = 1.4; sample in figure 12*b*). Similarly, biotite grains separated from the same rock sample show evidence of argon loss (figures 12*c*,*d*) and yield distinct plateau ages of 908 ± 2 Ma and 944 ± 6 Ma. We attribute these ages to thermal/chemical alteration effects of the c. 950 Ma intrusive event. These effects were apparently not widespread, because no evidence of resetting was found in the nearby Mukanda-Buhoro samples.

Discussion

The paleomagnetic pole for the late Kibaran intrusives at 17° S, 113° E (A95 = 7°) can be compared with similar age paleomagnetic poles from the Kalahari craton, Baltica, Laurentia, and Australia. Hoffman (1991) proposed that the Grenvillian-Irumide belts with ages of 1000–1100 Ma represent the sutures of Rodinia's final assembly, so at about 1200–1250 Ma we would expect some, or all, of the continents to still be dispersed.

Dopplehammer and Hargraves (1994) have calculated a mean c. 1200 Ma pole from the study of four kimberlite pipes in the Premier kimberlite swarm. The mean pole for these intrusions into the Kalahari Craton is 33° S, 233° E ($\alpha 95$ = 16°). The paleolatitudinal position of Baltica at about 1250 Ma is constrained by four poles with a mean of 1° S, 154° E (A95 = 6°; Pesonen et al. 1989). The Gardar Province of Greenland provides a well-dated 1250 Ma pole for Laurentia at 8° N, 224° E

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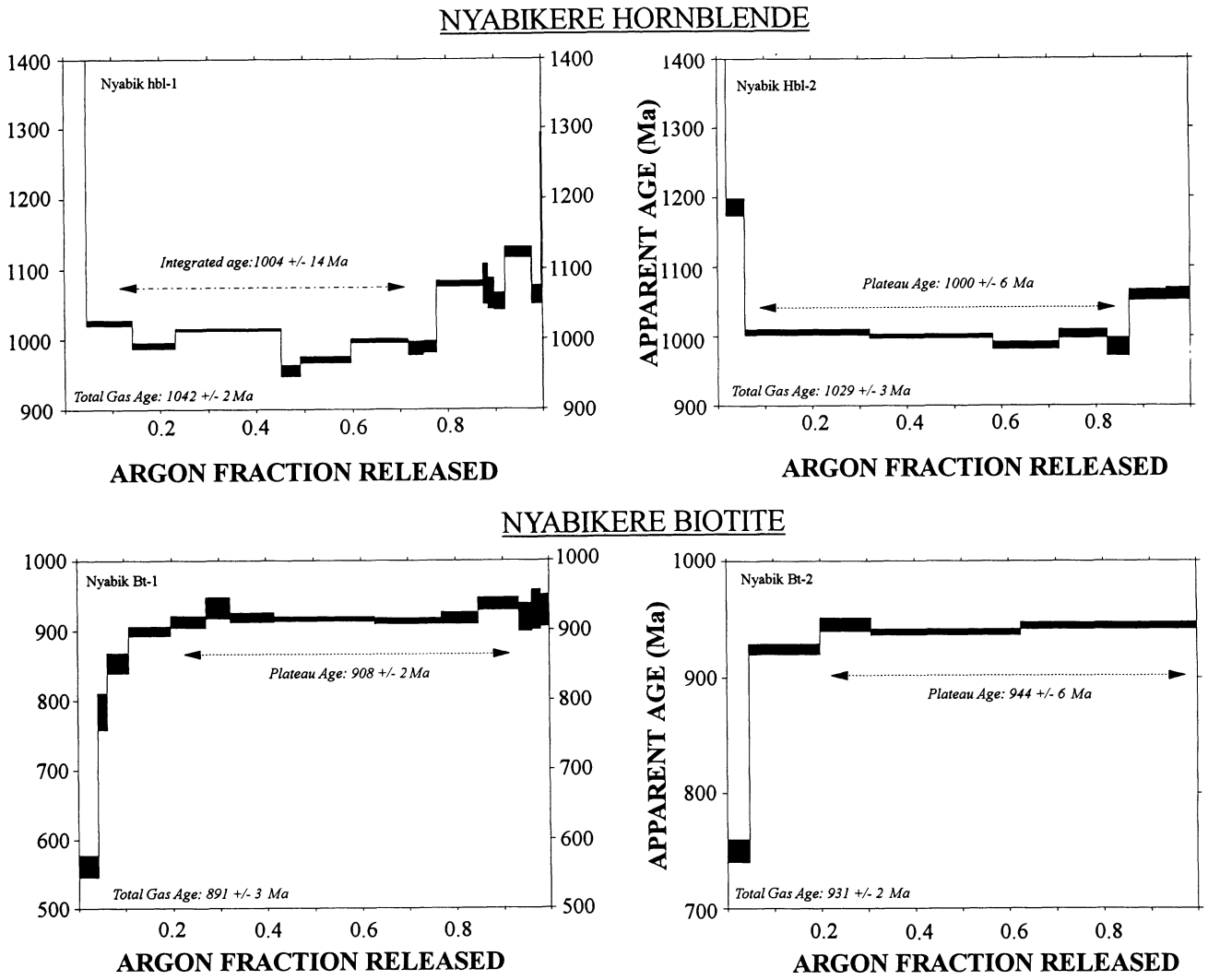


Figure 12. Typical argon release spectra for (*a,b*) hornblende splits from the Nyabikere intrusion (each split is 6–7 grains of hornblende); and (*c,d*) for individual biotite grains from the Nyabikere intrusion.

($\alpha 95 = 9°$; Piper 1992; Paslick et al. 1993). The paleomagnetic poles from Greenland are rotated to Laurentia according to Euler pole of Wells and Verhoogen (1967). A number of other similar age poles from North America and Greenland (Piper 1992; Abrahamsen and Van der Voo 1987; Stupavsky and Symons 1982; Fahrig 1971) are combined with the Gardar results to calculate a mean 1250 Ma pole for Laurentia at 13° N, 200° E (A95 = 11°). The paleolatitudinal position of Australia is only loosely constrained at 1250 Ma by the apparent polar wander path of Idnurm and Giddings (1988) with the pole at about 80° N, 003° E (A95 = 15°). These poles are plotted in figure 13*a*. Figure 13*b* shows these paleomagnetic poles rotated into the Piper (1987) configuration, and figure 13*c* shows the poles rotated into the Dalziel (1992) configuration. The Piper (1987) reconstruction requires that coeval poles must fall at the same point along his apparent polar wander path (APWP) when rotated to his configuration. We have superimposed the 1275–1175 Ma segment of this APWP on figure 13*b* and note that the Congo Craton pole from the Burundi rocks does not conform to his reconstruction. The Baltica and Kalahari means poles fall on, or near, his path but at younger ages. These data collectively suggest that his reconstruction may not be valid at 1250 Ma. When rotated into the configuration of Dalziel (1992), the scatter of these poles suggest that his configuration was not fully assembled at 1250 Ma as might be expected if assembly occurred at 1100 Ma. Hoffman (1991) does not supply any specific rotation parameters to assemble Rodinia into his configuration, and therefore it can be tested by simply reconstructing the continents according to their correct paleolatitudes

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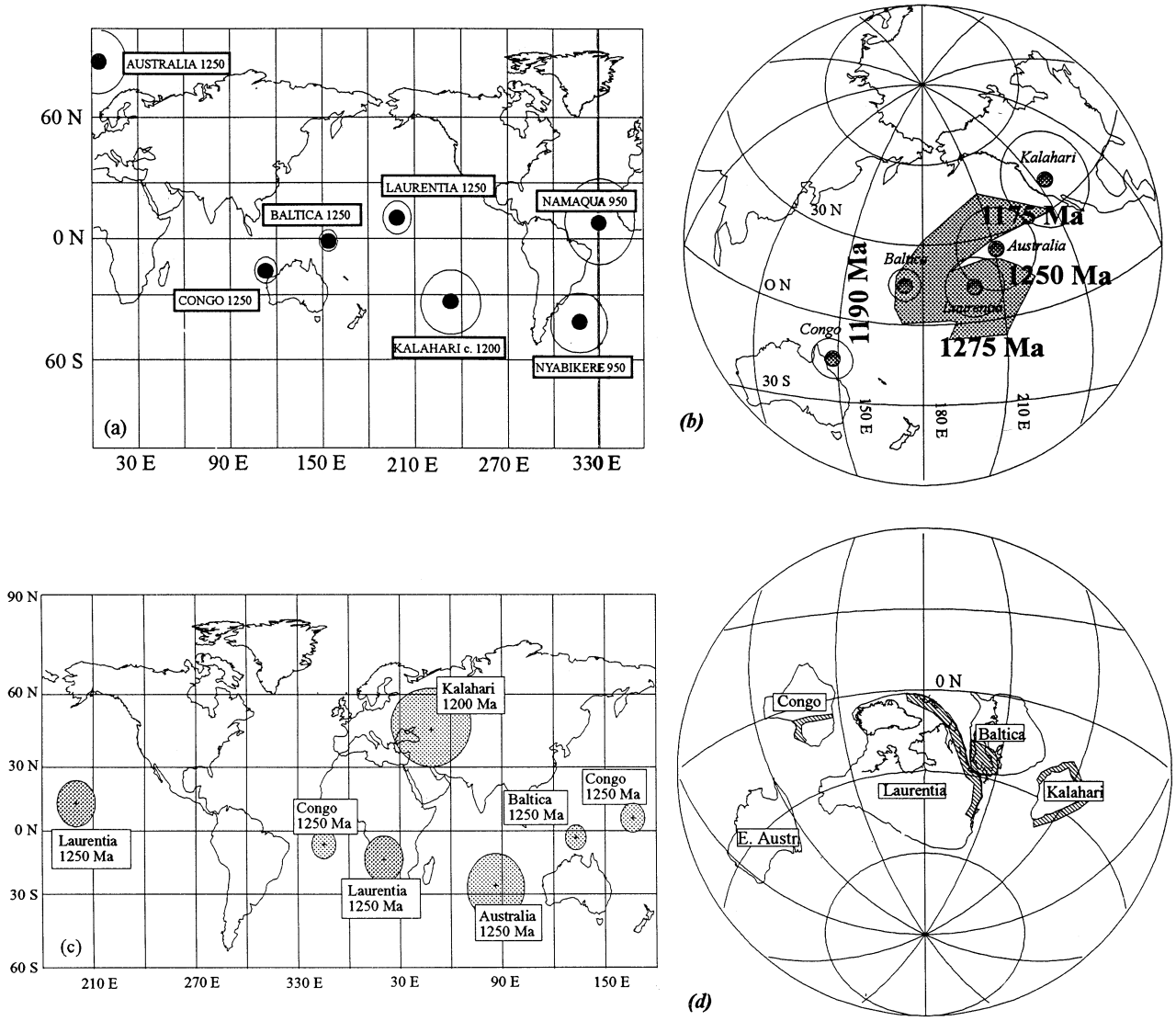


Figure 13. (*a*) Plot of 1200–1250 Ma paleomagnetic poles (present-day coordinates) for the Congo craton, Kalahari craton, Baltica, Laurentia, and Australia and c. 950 Ma poles from the Congo craton (Nyabikere) and Kalahari craton (Namaqua); (*b*) The same 1200–1250 Ma paleomagnetic poles as figure 13*a* rotated to the configuration of Piper (1987; Laurentia fixed). Note that the Congo craton pole does not fall on the path, and the poles from the Kalahari craton and Baltica fall on younger segments of the apparent polar wander path. (*c*) The figure 13*a* poles rotated to the Dalziel (1992; Laurentia fixed) configuration. The paleomagnetic poles for the Congo craton and Laurentia are shown both as poles and antipoles; and (*d*) a c. 1200–1250 Ma paleolatitudinal plot of the Congo craton, Kalahari craton, Baltica, Australia, and Laurentia based on the paleomagnetic poles listed in the text and positioned longitudinally to minimize the misfit of the Hoffman (1991) configuration.

and moving the continents longitudinally toward Laurentia. Figure 13*d* shows the paleolatitudinal position of these continents based on the paleomagnetic poles where the paleolongitudes are unconstrained. The Grenville-Kibaran-Irumide collisional belts, along which the constituent cratons of Rodinia were amalgamated, are shown for comparison in this reconstruction. The polarities of the poles were chosen so as to show minimal latitudinal separations of the continents with respect to Rodinia. The degrees of freedom allowed by the paleomagnetic data (i.e., no paleolongitudes, errors in pole position, and choice of polarity) show that it is possible that some of the cratons may have approached their ultimate Rodinia configuration by c. 1250 Ma.

The paleomagnetic pole for the Nyabikere massif at c. 950 Ma is 43° S, 317° E ($\alpha 95 = 14°$) must

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be considered preliminary at this point because it is based on a single site. It is distinct from a coeval mean pole for the Kalahari Craton using the average of Namaqua poles from Onstott et al. (1986) at 9° N, 331° E (A95 = 23°; figure 13*a*). Because of the uncertainties in these poles we cannot rigorously test the Dalziel (1992), Hoffman (1991), or Piper (1987) reconstructions.

Conclusions

The late Kibaran intrusives yield a key paleomagnetic pole for the Congo Craton at 17° S, 113° E (A95 = 7°) The 1236 ± 24 Ma age of magnetization is constrained by $^{40}Ar/^{39}Ar$ ages of mica samples from the Mukanda-Buhoro massif and an estimate of the cooling history using these ages and U-Pb ages from nearby, cogenetic intrusions. A comparison of paleomagnetic poles from Kalahari craton, Congo craton, Laurentia, Baltica, and Australia indicates that, while some of the continents may Have been close to their proposed Rodinia configuration, further tectonic displacements are required before the reconstruction of either Dalziel (1992) or Hoffman (1991) is valid. This observation is consistent with the hypothesis that final amalgamation of Rodinia occurred at 1100 Ma. The 1200–1250 Ma paleomagnetic poles from the Kalahari craton, Congo craton, and Baltica violate the supercontinent configuration envisioned by Piper (1987). This observation suggests that his reconstruction is not valid for this time interval and casts reasonable doubt on the existence of a long-lived supercontinent.

The Nyabikere massif is associated with the other 1200–1275 Ma late Kibaran mafic intrusions; however, the ca 950 Ma $^{40}Ar/^{39}Ar$ ages from hornblende and biotite suggest that the isotopic systems and remanent magnetization were reset by a later areally-limited thermal/chemical event.

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